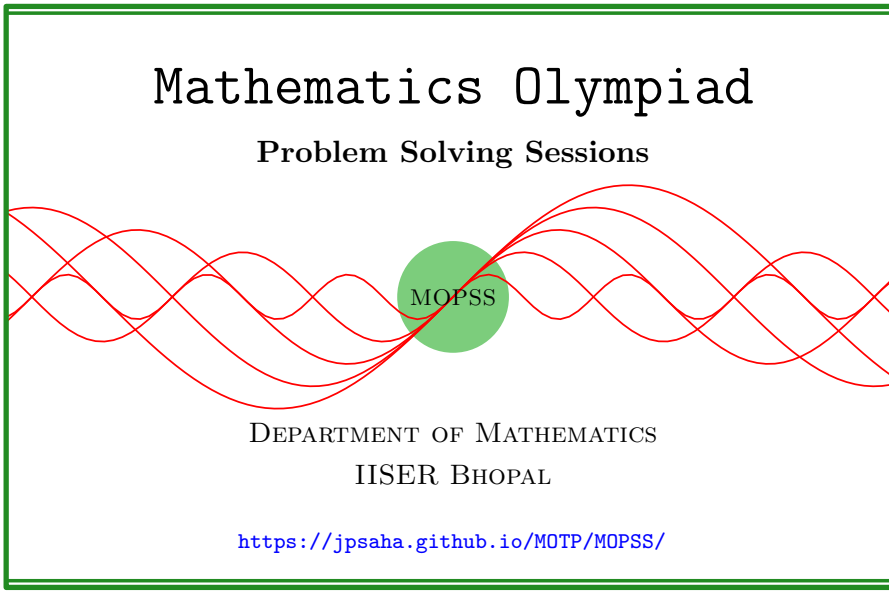
MOPSS

13 December 2025



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

Exercise 1 (Japan Mathematical Olympiad 1992 P1)**.** Let x, y be coprime positive integers with $xy > 1$, and let n be an even positive integer. Prove that $x^n + y^n$ is not divisible by $x + y$.

Walkthrough —

(a) Assume that $x + y$ divides $x^n + y^n$.

(b) Since x is congruent to $-y$ modulo $x + y$, and n is even, it follows that $x^n + y^n \equiv 2y^n \pmod{x + y}$.

(c) Conclude that $x + y$ divides 2.

Solution 1. On the contrary, let us assume that $x + y$ divides $x^n + y^n$. Since x is congruent to $-y$ modulo $x + y$, and n is even, it follows that x^n is congruent to y^n modulo $x + y$. This shows that $x^n + y^n$ is congruent to $2y^n$ modulo $x + y$. Therefore, $x + y$ divides $2y^n$. Since x, y are coprime, it follows that $x + y$ and y^n are coprime. Hence, $x + y$ divides 2. Since x, y are positive integers with $xy > 1$, it follows that $x + y \geq 3$, which is a contradiction. Thus, $x + y$ does not divide $x^n + y^n$. ■

Exercise 2 (Japan Mathematical Olympiad 1993 P2)**.** For a positive integer n, let $d(n)$ denote its largest odd divisor. Define

$$D(n) = d(1) + d(2) + \cdots + d(n),$$
$$T(n) = 1 + 2 + \cdots + n.$$

Show that there exist infinitely many positive integers n such that $3D(n) = 2T(n)$.

Walkthrough —

(a) Check that $n = 2, 6, 14, 30$ works.

(b) This may suggest that $n = 2^m - 2$ works for all $m \geq 2$.

(c) Does induction help?

Solution 2. Let us establish the following claim.

Claim — For any positive integer n,

$$D(2^n) = 2^n + \frac{1}{3}(2^n - 1)(2^n - 2)$$

holds.

Proof of the Claim. We prove the claim by induction on n. Note that

$$\begin{aligned} D(2) &= d(1) + d(2) \\ &= 1 + 1, \\ &= 2 + \frac{1}{3}(2-1)(2-2), \\ D(2^2) &= D(2) + d(3) + d(4) \\ &= 2 + 3 + 1 \\ &= 6 \\ &= 2^2 + \frac{1}{3}(2^2-1)(2^2-2) \end{aligned}$$

hold. Now, let $k \geq 2$ be a positive integer, and assume that the claim holds for for all positive integers up to k. Note that

$$\begin{aligned} & D(2^{k+1}) \\ &= D(2^k) + \sum_{i=1}^{2^k} d(2^k + i) \\ &= D(2^k) + \sum_{i=1}^{2^{k-1}} d(2^k + 2i - 1) + \sum_{i=1}^{2^{k-1}} d(2^k + 2i) \\ &= D(2^k) + \sum_{i=1}^{2^{k-1}} (2^k + 2i - 1) + \sum_{i=1}^{2^{k-1}} d(2^{k-1} + i) \\ &= D(2^k) + \frac{2^{k-1}}{2} \cdot (2^k + 1 + 2^k + 2 \cdot 2^{k-1} - 1) + D(2^k) - D(2^{k-1}) \\ &= 2D(2^k) - D(2^{k-1}) + 2^{k-2}(2^k + 2^{k+1}) \\ &= 2\left(2^k + \frac{1}{3}(2^k-1)(2^k-2)\right) \\ &\quad - \left(2^{k-1} + \frac{1}{3}(2^{k-1}-1)(2^{k-1}-2)\right) + 2^{k-2}(2^k + 2^{k+1}) \\ &= 2^{k+1} + \frac{1}{3}\left(2(2^k-1)(2^k-2) - (2^{k-1}-1)(2^{k-1}-2)\right) \\ &\quad + 2^{k-2}(2^k + 2^{k+1}) - 2^{k-1} \\ &= 2^{k+1} + \frac{1}{3}\left(2(2x-1)(2x-2) - (x-1)(x-2) + \frac{3}{2}x(2x+4x)\right) - x \\ &\quad (\text{where } x = 2^{k-1}) \\ &= 2^{k+1} + \frac{1}{3}\left(8x^2 - 12x + 4 - x^2 + 3x - 2 + 9x^2\right) - x \\ &= 2^{k+1} + \frac{1}{3}(16x^2 - 12x + 2) \end{aligned}$$

$$= 2^{k+1} + \frac{1}{3}(4x-1)(4x-2)$$
$$= 2^{k+1} + \frac{1}{3}(2^{k+1}-1)(2^{k+1}-2).$$

□

For any integer $m \geq 2$, using the claim, we have

$$\begin{aligned} D(2^m-2) &= D(2^m) - d(2^m-1) - d(2^m) \\ &= D(2^m) - (2^m-1) - 1 \\ &= 2^m + \frac{1}{3}(2^m-1)(2^m-2) - 2^m \\ &= \frac{1}{3}(2^m-1)(2^m-2), \end{aligned}$$

and consequently, we obtain

$$\begin{aligned} 3D(2^m-2) &= (2^m-1)(2^m-2) \\ &= 2(1+2+\cdots+(2^m-2)) \\ &= 2T(2^m-2). \end{aligned}$$

■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)